

**National Water Academy (NWA) Training Programme on
“Advanced Technologies for Water Resources Management”**

6 - 10 January 2025

**Interdisciplinary Centre for Water Research (ICWaR)
Indian Institute of Science, Bengaluru, Karnataka- 560 012**



Interdisciplinary Centre for Water Research (ICWaR), Indian Institute of Science (IISc), Bengaluru organised a one-week Mandatory Cadre Training Programme (MCTP) on “Advanced Technologies for Water Resources Management”, during **06 – 10 January, 2025**. The MCTP was sponsored by the National Water Academy (NWA) to train STS Officers of CWES (Group-A) Batch – IV Level-2. The training programme comprised of lectures and tutorials by experts, covering advanced topics in Remote Sensing and GIS, Reservoir Operation, Irrigation Water Management and Canal Automation, Urban and Riverine Flood Assessment and Mitigation, Climate Change Impact, Internet of Things (IoT) – Introduction and Applications: Water quality and Emerging Contaminants.

Venue for the Lectures and Tutorials: Lecture Hall, Interdisciplinary Centre for Water Research (ICWaR), Indian Institute of Science (IISc), Bengaluru

Overview of the event: The event consisted of 28 STS Officers of CWES (Group-A) Batch IV Level 2 including 16 Deputy Directors, 11 Executive Engineers and 1 Deputy Commissioner.

S.No.	Name & Designation (Shri)	Department
1	Vipin Kumar Singh	Executive Engineer, Betwa River Board (BRB), Jhansi
2	Jeeta Ram	Dy. Director, CA (Irrigation) DTE, CWC (Hq)
3	Mohammad Afaq	Executive Engineer, LYD, Agra
4	Rajesh Vellapat	Dy. Director, WPC
5	Saket Krishna	Dy. Director, M&A, Jaipur
6	Ashish Tripathi	Dy. Director, Monitoring, Gandhinagar
7	Saurabh Sharan	Dy. Director, DSR Directorate, HQ
8	Shobhit Chandra Misra	Executive Engineer, MGD-I, Lucknow
9	Vivek Kumar Singh	Dy. Director, MoFPI (on deputation)
10	Arkaprabha Majumder	Dy. Director, DVRRC, Maithon
11	Indrajeet Kumar	Dy. Director, CA (HWS&F) DTE
12	Raj Kumar	Executive Engineer, MGD-III, Varanasi

13	Aditya Moola	Dy. Director, Monitoring, Chennai
14	Abhijeet Kashliwal	Executive Engineer, UBD, Dibrugarh
15	Kumar Kushal	Executive Engineer, LGD-1 (MGD-IV), Patna
16	Chaitanya K.S	Dy. Director, NWA, Pune
17	Panneer Selvam L.	Executive Engineer, SRD, Coimbatore
18	Pushpendu Majumdar	Executive Engineer, Tapi Div, Surat
19	Manas Hans Yadav	Dy. Director, Monitoring, Agra
20	Harendra Singh	Executive Engineer, Chambal Div., Jaipur
21	Sarbjit Kumar Singh	Executive Engineer, MGD 2 Lucknow
22	Nagaraj V	Dy. Director, BCD (NW&S), CWC, New Delhi
23	Randhir Kumar Choudhary	Dy. Director, CMDD (N&W) DTE
24	D. Ganesh Kumar	Dy. Director, (Appraisal), Nagpur
25	Amit Mittal	Executive Engineer, Chenab Division, Jammu
26	Majeti Ravi	Dy. Director, Informally posted in PPA, Hyderabad
27	Sharad Kumar Kaushal	Deputy Commissioner, FM, D/o WR, RD & GR
28	Kailash Kisanaji Lakhe	Dy. Director, WSE DTE

Faculty members from Indian Institute of Science: Prof. Praveen C. Ramamurthy, Prof. M. Sekhar, Prof. V. V. Srinivas, Prof. L. N. Rao Dr. Rajarshi Das Bhowmik, Dr. Bramha Dutt Vishwakarma, Dr. Debsunder Dutta, Dr. Soumita Boral and Prof. K.S. Kasiviswanathan (invited lecture- IIT Roorkee) delivered lectures on different topics related to advanced technologies for water resources management.

Each day of the training programme was divided into two sessions. In the morning sessions, two lectures were conducted and in the afternoon sessions: tutorials on the topics covered were given. A visit to Karnataka State Natural Disaster Monitoring Centre (KSNDMC) was organized on 8th January, 2025 in the afternoon session of the training programme. All the officers visited KSNDMC where the officials described and demonstrated to the delegates the monitoring mechanism and early warning systems of natural disasters in Karnataka, weather monitoring instruments and Varuna Mitra a 24/7 helpline service that provides weather forecasts to farmers and the public in Karnataka.



Topic-wise distribution

Faculty	Topic
Dr. Debsunder Dutta (DD)	Introduction to remote sensing, GIS with applications to water resources planning and management, water resources assessment, surface energy balance.
Dr. Bramha Dutt Vishwakarma (BDV)	Geodetic remote sensing, water balance, drought and flood monitoring, introduction to DEM – with applications to water resources planning and management.
Prof K. S. Kasiviswanathan (KSK) (IIT Roorkee, invited lecture)	Reservoir operation-systems techniques; computer modelling; optimization; simulation of multi-reservoir system operation.
Prof. Sekhar Muddu (SM)	Irrigation water management - water demand assessment, crop-water budget, groundwater resource and irrigation, smart-irrigation, soil moisture.
Prof. V. V. Srinivas (VVS)	Urban flooding: introduction to urban hydrology; IDF and DAD relationships; design hyetographs, riverine flood assessment-introduction and approaches.
Dr. Rajarshi Das Bhowmik (RDB)	Watershed modelling, canal automation, climate change impacts, structural and non-structural measures to mitigate flood risk.
Prof. Praveen C. Ramamurthy (PCR), Dr. Soumita Boral (SB) and Prof. L. N. Rao (LNR)	Internet of Things (IoT)-introduction and applications: water quality, Emerging contaminants and measurement techniques. Importance of water quality, link with SDGs and the broader picture. Revisiting water quality: spatio-temporal variations, ambient background concentration, interlinks between parameters, novel contaminants, role of sediment, organic matter, residence time.